

COMPREHENSIVE PRODUCT RANGE • MASTER E-CATALOGUE

Industrial Compressed Air & Process Thermal Engineering

High-Efficiency Refrigerated & Desiccant Air Dryers, Industrial Process Chillers, Specialized Anodizing/Medical Chillers, FRP Cooling Towers, Pressure Vessels & Filtration Systems



1,200+

Industrial Installations

18+

Years Engineering
Excellence

10

Core Product Lines

100%

FAT Proof Tested

ENGINEERING LEADERSHIP SINCE 2008

About Win Equipments

Pioneering precision compressed air treatment & process thermal systems in Coimbatore, India

Founded in 2008 by industrial engineering veteran **Mr. Ramasamy Ananthakumar**, Win Equipments has grown from an innovative design workshop into one of South India's foremost manufacturers of high-efficiency industrial compressed air dryers, process chillers, cooling towers, and heavy pressure vessels.

Operating from its integrated manufacturing plant at **SF No: 4, 195 B, Kallangadu, Arasur Post, Coimbatore**, the company upholds a foundational commitment encapsulated in its guiding motto: *"Save Water and Power"*. Every machine is engineered specifically to withstand severe Indian tropical and Middle-Eastern ambient conditions (up to 45°C continuous and 50°C peak), delivering unyielding reliability where generic machinery fails.



Win Equipments Manufacturing Plant, Arasur Works, Coimbatore



Integrated Design & Fabrication

In-house CNC machining, automated submerged-arc welding (SAW), high-precision sheet metal shearing, and electrostatic powder coating bays ensure total quality control from raw steel plates to final crated skids.



Extreme Tropical Ambient Duty

Unlike European and imported machinery that trips on high head pressure during Indian summer heat peaks, Win Equipments oversized condenser coils and generous heat exchanger areas deliver full rated capacity at 45°C ambient.



Thermodynamic Energy Conservation

From zero-air-loss capacitive drains saving ₹42,000/yr per compressor, to 3-in-1 modular regenerator heat exchangers reducing refrigeration kW by 40%, our systems are engineered to minimize client utility bills.



Direct Factory Engineering Support

Over 1,200 active installations across India backed by direct Coimbatore factory field engineers, fully stocked OEM spare parts warehousing, and comprehensive Annual Maintenance Contracts (AMC).

INFRASTRUCTURE & CERTIFIED QUALITY

Manufacturing Infrastructure & Rigorous Quality Testing

Accredited quality management systems conforming to ISO 9001:2015, IAF & DAC standards



✓ 4-Stage Factory Acceptance Testing (FAT)

- **Hydrostatic Proof Testing:** 1.5× design pressure hold test on all pressure vessels, coils, and buffer tanks.
- **Helium Leak Detection:** Mass spectrometer vacuum testing of refrigeration circuits (<0.1 g/year loss).
- **Thermal Calorimeter Testing:** 8-hour continuous full-load simulation verifying kW heat removal and dew point.
- **Electrical Safety & Megger Testing:** High-potential dielectric breakdown and earthing bonding audit.

★ Global Accreditation Badges

- **ISO 9001:2015:** Certified Quality Management System covering design, manufacturing, testing, and servicing.
- **IAF (International Accreditation Forum):** Globally recognized conformity assessment and trade compliance.
- **DAC (Dubai Accreditation Center):** Certified for extreme desert high-ambient installations across GCC regions.
- **Boiler Regulations:** Fabrication compliance with IS 2825 Class II and ASME Section VIII Division 1.

Zero-Defect Quality Policy

Every system manufactured at Arasur Works undergoes individualized FAT protocol testing. Test data, refrigerant charge records, electrical logs, and welder qualification credentials are permanently archived and supplied with each machine's documentation package.

COMPRESSED AIR TREATMENT SYSTEMS

High-Efficiency Refrigerated Air Dryers

Non-Cycling Direct-Expansion Technology Delivering Constant +3°C Pressure Dew Point Under Extreme Tropical Ambient Conditions



Win Equipments WRD Series refrigerated compressed air dryers remove moisture vapor from compressed air networks through precision thermodynamic chilling. Engineered specifically for Indian and GCC tropical environments with ambient temperatures reaching 45°C and air inlet temperatures up to 50°C, WRD dryers protect sensitive pneumatic machinery, CNC spindles, and automated paint lines from liquid water, emulsion sludge, and corrosion.

Working Principle: Saturated compressed air enters the high-efficiency 3-in-1 modular aluminum heat exchanger. In the air-to-air regenerator stage, incoming warm wet air is precooled by outgoing cold...

Design Ratings: Nominal Working Pressure: 7.0 bar g (Max 16.0 bar g) | Max Inlet Temp: 50°C | Ambient Temp: 35°C | Pressure Dew Point: +3°C | Refrigerant: R134a / R407C

MODEL CODE	NOMINAL CFM	FLOW M ³ /HR	POWER KW	SUPPLY VOLTAGE	IN/OUT CONN	DIMENSIONS (L×W×H MM)
WRD 20 S	20 CFM	34	0.45 kW	230V / 1Ph / 50Hz	1/2" BSP	450 × 450 × 550
WRD 30 S	30 CFM	51	0.45 kW	230V / 1Ph / 50Hz	1/2" BSP	450 × 450 × 550
WRD 40 S	40 CFM	68	0.45 kW	230V / 1Ph / 50Hz	1/2" BSP	450 × 450 × 550
WRD 60 S	60 CFM	102	0.65 kW	230V / 1Ph / 50Hz	1" BSP	500 × 550 × 750
WRD 80 S	80 CFM	136	0.65 kW	230V / 1Ph / 50Hz	1" BSP	500 × 550 × 750
WRD 100 S	100 CFM	170	0.85 kW	230V / 1Ph / 50Hz	1" BSP	500 × 550 × 750

Key Engineering Advantages

- **3-in-1 Monoblock Aluminum Heat Exchanger:** Combines air-to-air precooling/reheating, direct-expansion evaporation, and high-velocity centrifugal water separation.
- **Mechanical Hot Gas Bypass Modulation:** Features pressure-balanced hot gas bypass injection that automatically modulates refrigerant flow for optimal efficiency.

Primary Applications

- **Textile Spinning & Weaving:** Protects electronic yarn clearers, autoconers, and air-jet loom nozzles from moisture.
- **CNC Machine Tooling & Automotive:** Eliminates moisture-induced solenoid sticking, spindle taper oxidation, and pneumatic issues.
- **Automated Spray Painting:** Prevents micro-blistering, cratering, and paint fish-eye defects in robotic automotive applications.

CRITICAL CLEAN AIR TREATMENT SYSTEMS

Heatless Twin-Tower Desiccant Air Dryers

Pressure Swing Adsorption (PSA) Technology Delivering Guaranteed -40°C to -70°C Pressure Dew Point for Ultra-Critical Industrial Applications



Win Equipments WHD Series twin-tower desiccant air dryers eliminate atmospheric moisture vapor to sub-zero dew points. Designed for critical manufacturing where any trace of moisture causes irreversible process failures—such as pharmaceutical formulation, semiconductor fabrication, and sub-zero pneumatic instrumentation—WHD dryers utilize Pressure Swing Adsorption (PSA) to guarantee -40°C pressure dew point standard (with -70°C optional using 13X molecular sieve).

Working Principle: Compressed air enters through dual angle-seat non-lubricated switching valves and flows upward through the online adsorbing desiccant bed. Microscopic pores in the spherical activa...

Design Ratings: Operating Pressure: 7.0 to 16.0 bar g | Inlet Air Temp: 35°C to 45°C | Pressure Dew Point: -40°C Standard (-70°C Optional) | Cycle Time: 8 min (4 min Adsorption / 4 min Regeneration) | Vessel Code: IS 2825 / ASME Sec VIII Div 1

MODEL CODE	NOMINAL CFM	FLOW M ³ /HR	RATED PRESSURE	FLANGE CONN	CONTROL SUPPLY	DIMENSIONS (L×W×H MM)
WHD - 030	300 CFM	510	7–16 bar g	1.5" NB Flange	220V / 1Ph / 50Hz	1450 × 1100 × 2500
WHD - 040	400 CFM	680	7–16 bar g	2" NB Flange	220V / 1Ph / 50Hz	1450 × 1200 × 2500
WHD - 050	500 CFM	850	7–16 bar g	3" NB Flange	220V / 1Ph / 50Hz	1650 × 1250 × 2300
WHD - 060	600 CFM	1,020	7–16 bar g	3" NB Flange	220V / 1Ph / 50Hz	1650 × 1250 × 2500
WHD - 075	750 CFM	1,275	7–16 bar g	3" NB Flange	220V / 1Ph / 50Hz	1800 × 1300 × 2600
WHD - 100	1,000 CFM	1,700	7–16 bar g	4" NB Flange	220V / 1Ph / 50Hz	2000 × 1400 × 2700

Key Engineering Advantages

- **High Crush-Strength Spherical Activated Alumina:** Premium 4–8mm spherical desiccant with exceptional surface area (>340 m²/g) and high mechanical attr...
- **High-Cycle Angle-Seat Pneumatic Valves:** Piston-actuated stainless steel angle seat switching valves tested to over 1,000,000 continuous swit...

Primary Applications

- **Pharmaceutical & Bulk API Manufacturing:** Guarantees moisture-free sterile air for pill coating, fluidized bed drying, powder c...
- **Automotive Robotics & High-Gloss Finish:** Delivers bone-dry propellant air for automated electrostatic bells, preventing water-...
- **Sub-Zero Outdoor Air Distribution:** Eliminates catastrophic pipe freeze-up in exposed industrial pipelines during winter ...

PROCESS THERMAL MANAGEMENT SYSTEMS

High-Precision Industrial Process Chillers

Packaged Closed-Loop Air & Water-Cooled Chillers with Integrated SS304 Buffer Reservoir & Stainless Centrifugal Circulation Pumps



Win Equipments WCP Series industrial process chillers provide closed-loop liquid thermal management for factory automation, plastic molding, laser cutting, and chemical processing. Built to withstand continuous 24/7 industrial duty in high-ambient environments up to 45°C, WCP chillers integrate high-efficiency scroll compressors, a heavy-gauge insulated SS304 buffer reservoir, and an all-stainless centrifugal circulating pump within a compact powder-coated steel monoblock skid.

Working Principle: Process fluid absorbs heat from industrial machinery and returns to the internal SS304 insulated buffer tank. A stainless centrifugal pump draws fluid from the tank and circulates ...

Design Ratings: Leaving Water Temp: +12°C (Range +5°C to +25°C) | Water Delta T: 5°C | Ambient Temp: 35°C (Rated up to 45°C) | Refrigerant: R410A / R407C Eco-friendly | Buffer Tank: SS304 Insulated

MODEL CODE	NOMINAL TR	COOLING KW	COMP POWER KW	SUPPLY VOLTAGE	WATER FLOW LPM	PUMP HP
WCP 005	0.5 TR	1.7 kW	0.9 kW	230V / 1Ph / 50Hz	6 LPM	0.5 HP
WCP 010	1.0 TR	3.5 kW	1.5 kW	230V / 1Ph / 50Hz	10 LPM	0.5 HP
WCP 020	2.0 TR	7.0 kW	2.5 kW	230V / 1Ph / 50Hz	21 LPM	0.75 HP
WCP 030	3.0 TR	10.5 kW	3.4 kW	415V / 3Ph / 50Hz	32 LPM	1.0 HP
WCP 050	5.0 TR	17.5 kW	6.0 kW	415V / 3Ph / 50Hz	52 LPM	1.5 HP
WCP 075	7.5 TR	26.3 kW	8.5 kW	415V / 3Ph / 50Hz	78 LPM	2.0 HP

Key Engineering Advantages

- **All-Stainless SS304 Non-Ferrous Wetted Loop:** Internal buffer tank, pump impellers, and brazed plate heat exchanger are manufactured from stainless...
- **High-Efficiency Copeland / Danfoss Scroll Compressors:** Utilizes hermetic scroll compressors with 70% fewer moving parts than reciprocating machines. Delive...

Primary Applications

- **Plastic Injection Molding & Blow Extrusion:** Cuts mold cycle times by up to 28% through rapid heat extraction and prevents part wa...
- **Fiber Laser Cutting & Welding:** Maintains resonator and cutting head optics at exact temperatures, preventing thermal...
- **CNC Electro-Spindles & Wire EDM:** Extracts motorized spindle stator heat and stabilizes dielectric deionized water bath...

SEVERE CHEMICAL & HEALTHCARE THERMAL ENGINEERING

Specialized Application Process Chillers

Custom Titanium Anodizing Chillers, N+1 Redundant Medical Scan Chillers & Severe Chemical Acid Cooling Systems



Standard commercial chillers fail rapidly when exposed to aggressive acid electrolytes or when supporting zero-downtime medical imaging magnets. Win Equipments specializes in custom-engineered thermal management packages: WAN Series Titanium Anodizing Chillers resisting 20% sulfuric acid baths; WMS Series Medical Scan Chillers providing dual-redundant cooling for clinical MRI/CT magnets; and WAC Series Acid Process Chillers featuring Hastelloy C-276 metallurgy and sealless magnetic drive pumps.

Working Principle: WAN Anodizing Chillers utilize Grade 2 pure Titanium seamless immersion coils or external Titanium plate heat exchangers to absorb intense exothermic rectifier heat, holding bath t...

Design Ratings: WAN Anodizing: Bath +18°C to +21°C (Hardcoat -2°C to +4°C) | WMS Medical: Supply +8°C to +15°C (±0.2°C) | WAC Acid: Acid bath -5°C to +15°C | Exchanger: Titanium Gr. 2 / Hastelloy / PTFE

MODEL CODE	SERIES / APP	CAPACITY TR	COOLING KW	RECTIFIER / DUTY	WETTED METALLURGY	COMPRESSOR SETUP
WAN 020	WAN Anodizing	2.0 TR	7.0 kW	Up to 350 A	Pure Titanium Gr. 2	Copeland Scroll
WAN 050	WAN Anodizing	5.0 TR	17.5 kW	Up to 1,000 A	Pure Titanium Gr. 2	Copeland Scroll
WAN 100	WAN Anodizing	10.0 TR	35.1 kW	Up to 2,000 A	Pure Titanium Gr. 2	Copeland / Danfoss
WAN 200	WAN Anodizing	20.0 TR	70.3 kW	Up to 4,000 A	Pure Titanium Gr. 2	Dual Scroll Circuits
WAN 500	WAN Anodizing	50.0 TR	175.8 kW	Up to 10,000 A	Pure Titanium Shell & Tube	Bitzer Semi-Hermetic
WAN 1000	WAN Anodizing	100.0 TR	351.6 kW	Up to 20,000 A	Pure Titanium Shell & Tube	Twin Semi-Herm / Screw

Key Engineering Advantages

- **Grade 2 Pure Titanium Acid Immersion Metallurgy:** Features seamless Grade 2 Titanium heat exchanger coils fabricated with inert gas TIG welding. Compl...
- **Dual Redundant (N+1) Circuits & Auto City Bypass:** WMS Series incorporates two completely separate refrigeration compressors, independent condensers, a...

Primary Applications

- **Architectural & Aerospace Anodizing:** Maintains bath temperatures within ±0.5°C during high-amperage cycles, ensuring unifo...
- **Hospital Diagnostic Imaging (MRI / CT / PET):** Protects superconducting magnets from cryogenic helium boil-off with dual 100% redund...
- **Lead-Acid & Lithium Battery Formation:** Dissipates intense exothermic heat generated during high-rate battery charging, preve...

COMMERCIAL & INDUSTRIAL ICE PLANTS

Heavy-Duty Industrial Ice Flake Machines

Continuous Production of Sub-Cooled Dry Ice Flakes (-5°C to -8°C) with Stationary Vertical SUS304 Evaporator Drums & Bitzer Compressors



Win Equipments WFI Series industrial ice flake machines produce continuous, sub-cooled dry ice flakes engineered for commercial fisheries, poultry processing, concrete dam pouring, and chemical synthesis. Unlike conventional rotating-drum ice machines that suffer from refrigerant seal leaks, the WFI Series employs a stationary vertical SUS304 evaporator drum with an internal rotating spiral ice blade, delivering zero refrigerant leakage, ultra-low maintenance, and maximum thermodynamic efficiency.

Working Principle: Fresh or seawater is pumped from a lower stainless steel basin into an overhead rotary water distributing pan that coats the inner vertical wall of the stationary evaporator drum w...

Design Ratings: Ice Production Basis: Water Inlet +16°C | Ambient Temp +25°C | Evaporation Temp -22°C | Condensing Temp +40°C | Flake Temp: -5°C to -8°C | Flake Thickness: 1.5 to 2.2 mm | Drum: Food Grade SUS304

MODEL CODE	DAILY OUTPUT TPD	DAILY OUTPUT KG/24H	POWER KW	COMPRESSOR MAKE & HP	CONDENSER TYPE	SKID DIMENSIONS (L×W×H MM)
WFI 010	1 TPD	1,000 kg	4.8 kW	Danfoss / Bitzer (5 HP)	Air / Water Cooled	1350 × 1050 × 980
WFI 020	2 TPD	2,000 kg	8.5 kW	Bitzer Semi-Herm (8.5 HP)	Water Cooled Shell & Tube	1550 × 1200 × 1150
WFI 030	3 TPD	3,000 kg	12.5 kW	Bitzer Semi-Herm (12 HP)	Water Cooled Shell & Tube	1800 × 1350 × 1300
WFI 050	5 TPD	5,000 kg	19.5 kW	Bitzer Semi-Herm (20 HP)	Water Cooled Shell & Tube	2100 × 1500 × 1550
WFI 100	10 TPD	10,000 kg	38.0 kW	Bitzer Semi-Herm (40 HP)	Water Cooled Shell & Tube	2600 × 1800 × 1850
WFI 150	15 TPD	15,000 kg	56.0 kW	Bitzer Screw (60 HP)	Water Cooled Shell & Tube	3100 × 2000 × 2100

Key Engineering Advantages

- **Stationary Vertical SUS304 Evaporator Drum:** The evaporator drum remains completely stationary while only the internal scraper blade rotates. Eli...
- **Sub-Cooled, Flat & Edueless Flake Geometry:** Produces thin (1.5–2.2 mm) sub-cooled dry flakes with zero free water. Their smooth, rounded edges p...

Primary Applications

- **Commercial Fisheries & Deep-Sea Trawlers:** Cools freshly caught seafood instantly without bruising fish flesh, preserving freshn...
- **Poultry Processing & Meat Abattoirs:** Maintains eviscerated carcasses at safe bacteriological temperatures (<4°C) during sp...
- **Mass Concrete Cooling for Dams & Highways:** Replaces mixing water with dry ice flakes to dissipate the intense heat of hydration,...

EVAPORATIVE HEAT REJECTION SYSTEMS

Industrial FRP Round & Square Cooling Towers

Aerodynamic Induced-Draft Counterflow & Crossflow Towers Engineered with Heavy-Duty Isophthalic FRP & High-Efficiency PVC Fills



Win Equipments WCT Series cooling towers provide evaporative heat rejection for industrial water chillers, plastic injection machines, diesel generator jackets, induction furnaces, and chemical plants. Available in two distinct architectural configurations—WCT-RL aerodynamic round bottle towers and WCT-SL modular square crossflow towers—each unit is constructed with premium isophthalic polyester resin reinforced with chopped strand fiberglass mats, ensuring complete resistance against corrosion, algae, and UV radiation.

Working Principle: Warm process water enters the tower and is distributed over high-efficiency virgin PVC honeycomb fills. In WCT-RL round towers, water is discharged through a calibrated multi-arm

Design Ratings: Nominal Flow: 0.6 m³/hr per TR (3.0 GPM/TR) | Hot Water In: 37°C | Cold Water Out: 32°C | Wet Bulb Temp: 28°C (Approach 4°C) | Casing: UV-Resistant FRP | Fill: PVC Cross-Flute

MODEL CODE	NOMINAL TR	WATER FLOW M ³ /HR	MOTOR HP	FAN RPM	ROUND DIA × H (MM)	SQUARE L×W×H (MM)
WCT 010	10 TR	6 m ³ /hr	0.5 HP	1440 RPM	Ø 930 × 1690	930 × 930 × 1690
WCT 015	15 TR	9 m ³ /hr	0.5 HP	1440 RPM	Ø 1050 × 1700	1050 × 1050 × 1700
WCT 020	20 TR	12 m ³ /hr	0.75 HP	1440 RPM	Ø 1150 × 1750	1150 × 1150 × 1750
WCT 025	25 TR	15 m ³ /hr	0.75 HP	1440 RPM	Ø 1250 × 1750	1250 × 1250 × 1750
WCT 030	30 TR	18 m ³ /hr	1.0 HP	960 RPM	Ø 1350 × 1850	1350 × 1350 × 1850
WCT 040	40 TR	24 m ³ /hr	1.0 HP	960 RPM	Ø 1500 × 1950	1500 × 1500 × 1950

Key Engineering Advantages

- **Heavy-Duty Isophthalic FRP Casing:** Precision molded with continuous chopped strand fiberglass and UV-inhibited isophthalic gelcoat. Imp...
- **Self-Rotating Alloy Sprinkler System:** Precision machined rotary sprinkler head rotates automatically on sealed stainless steel ball bearin...

Primary Applications

- **Plastic Processing & Blow Molding:** Cools hydraulic oil exchangers, barrel throat jackets, and chiller condensers for rel...
- **Steel Rolling & Induction Melting:** Protects water-cooled power cables, crucible induction coils, and SCR thyristor panel...
- **HVAC Central Water-Cooled Chillers:** Rejects building heat from centrifugal and screw chiller condensers at lowest electri...

CRITICAL CLOSED LOOP HEAT REJECTION

Closed Circuit Evaporative Coil Cooling Towers

Indirect Evaporative Cooling Isolating Process Fluid Inside Sealed Serpentine Heat Exchange Coils for 100% Contamination-Free Cooling



Win Equipments WCC Series closed circuit cooling towers provide indirect evaporative cooling for critical applications where process fluid cannot tolerate atmospheric air contamination, dust fouling, biological growth, or mineral scaling. By circulating the process fluid inside a completely sealed, continuous serpentine tube coil while an external deluge water spray and fan provide evaporative cooling, WCC towers safeguard expensive induction furnaces, laser resonators, and vacuum furnaces from catastrophic scale fouling.

Working Principle: Process fluid (clean water, deionized water, or glycol) circulates entirely inside a continuous multi-circuit serpentine coil bundle. Simultaneously, a secondary recirculating pump...

Design Ratings: Process Fluid In: 37°C | Process Fluid Out: 32°C | Wet Bulb Temp: 28°C | Coil Test Pressure: 25 bar g (Air under water) | Casing: Heavy FRP / HDG Panels | Spray Circuit: Low-Head Centrifugal Pump

MODEL CODE	CAPACITY TR	FAN MOTORS	FAN RPM	IN/OUT CONN	COIL VOL LITERS	DIMENSIONS (L×W×H MM)
WCC 40	40 TR	3.0 HP × 2	960 RPM	3" / 3"	72 L	3300 × 1500 × 1400
WCC 50	50 TR	3.0 HP × 2	960 RPM	3" / 3"	90 L	3300 × 1700 × 1400
WCC 75	75 TR	3.0 HP × 2	960 RPM	4" / 4"	135 L	3300 × 2000 × 1400
WCC 100	100 TR	5.0 HP × 2	960 RPM	4" / 4"	180 L	3600 × 2400 × 1600
WCC 150	150 TR	7.5 HP × 2	720 RPM	6" / 6"	270 L	4200 × 2800 × 1800

Key Engineering Advantages

- **Completely Sealed Contamination-Free Loop:** Eliminates scaling, airborne dust accumulation, bacterial slime, and oxygen ingestion inside client ...
- **Continuous Serpentine Tube Bundle Architecture:** Coils are manufactured from continuous smooth-bend heavy-gauge steel tubing, hot-dip galvanized after...

Primary Applications

- **Induction Melting & Billet Heating Furnaces:** Protects copper power coils and IGBT power supplies from mineral scale calcification ...
- **Continuous Casting & Vacuum Furnaces:** Delivers scale-free cooling water to vacuum chamber double-walls and turbomolecular p...
- **Industrial Air Compressor Jackets:** Replaces open tower cooling on oil-free screw compressor jackets, eliminating cylinde...

PRESSURE VESSEL & PNEUMATIC STORAGE SYSTEMS

Industrial Compressed Air Receiver Tanks

Heavy-Duty Carbon & Stainless Steel Pressure Vessels Fabricated to IS 2825 & ASME Section VIII Standards for Pulsation Dampening & Bulk Moisture Separation



Win Equipments WRV Series industrial air receiver tanks act as the vital pneumatic energy reservoir and primary moisture knockout vessel in compressed air installations. Fabricated in vertical and horizontal configurations from boiler-grade SA 516 Gr. 70 carbon steel (or SUS304 stainless steel), WRV vessels dampen pressure pulsations from reciprocating and screw compressors, satisfy sudden high-volume intermittent demand peaks, and eliminate compressor rapid cycling to save significant electrical power.

Working Principle: High-velocity discharge air from air compressors enters the bottom nozzle of the vertical receiver tank. The sudden drop in air velocity upon entering the large vessel volume cause...

Design Ratings: Design Codes: IS 2825 Class II / ASME Sec VIII Div 1 | Working Pressures: 7.0, 10.0, 16.0, and 40.0 bar g | Hydrostatic Test Pressure: 1.5× Design Pressure | Material: SA 516 Gr. 70 Boiler Plate | Finish: SA 2.5 Grit Blast + Epoxy + PU

MODEL CODE	CAPACITY LITERS	ORIENTATION	WORKING BAR	SHELL DIA MM	OVERALL H/L MM	IN/OUT FLANGE
WRV 025	250 L	Vertical	10 bar g	Ø 500 mm	1,550 mm	1" BSP
WRV 050	500 L	Vertical	10 bar g	Ø 650 mm	1,850 mm	1.5" BSP
WRV 100	1,000 L	Vertical	10 bar g	Ø 800 mm	2,250 mm	2" BSP / Flange
WRV 150	1,500 L	Vertical	10 bar g	Ø 950 mm	2,450 mm	2" Flange
WRV 200	2,000 L	Vertical	10 bar g	Ø 1,050 mm	2,650 mm	3" Flange
WRV 300	3,000 L	Vertical	10 bar g	Ø 1,250 mm	2,950 mm	3" Flange

Key Engineering Advantages

- **Boiler Quality SA 516 Grade 70 Fabrication:** Constructed from high-tensile boiler-quality carbon steel plates (SA 516 Gr. 70) with dished ends fo...
- **Full Code Compliance (IS 2825 / ASME Sec VIII):** Engineered and manufactured strictly in accordance with Indian Boiler Regulations (IS 2825 Class II/...

Primary Applications

- **Manufacturing Plant Central Ring Mains:** Eliminates network pressure drop and provides instantaneous compressed air storage du...
- **PET Stretch Blow Molding (40 bar):** Buffers high-pressure air demand during rapid mold bottle blowing, preventing compres...
- **Pneumatic Sandblasting & Shot Peening:** Maintains constant high-velocity nozzle pressure required for uniform surface cleanin...

PRECISION IN-LINE CONTAMINANT SEPARATION

High-Efficiency In-Line Compressed Air Filters

Four-Stage Progressive Coalescing & Particulate Filtration Delivering ISO 8573-1 Class 1 Air Purity Down to 0.01 Micron & 0.003 ppm Oil Content



Win Equipments WMF Series in-line compressed air filters remove solid particulates, liquid water aerosols, and oil mist from industrial air networks. Available in four progressive filtration grades—Grade P (pre-filter), Grade X (general purpose), Grade Y (micro-coalescer), and Grade A (activated carbon vapor absorber)—WMF filters ensure compliance with ISO 8573-1 air purity standards, protecting expensive automated machinery, instrumentation, and food packaging from contamination.

Working Principle: Compressed air enters the die-cast aluminum housing where internal aerodynamic flow channels guide air through multi-layered borosilicate microfiber filtration sleeves. Large parti...

Design Ratings: Max Working Pressure: 16.0 bar g | Max Working Temp: 65°C (Grade A: 30°C) | Housing Material: Alocrom Treated Die-Cast Aluminum | Element Media: Hydrophobic Borosilicate Microfiber

MODEL CODE	NOMINAL CFM	FLOW M ³ /HR	PIPE CONN	GRADE OPTIONS	DIMENSIONS (L×W×H MM)	WEIGHT KG	Standards
WMF 004	20 CFM	34	1/2" BSP	P, X, Y, A	90 × 90 × 220	1.1 kg	
WMF 008	45 CFM	76	1/2" BSP	P, X, Y, A	90 × 90 × 260	1.3 kg	
WMF 015	85 CFM	144	3/4" BSP	P, X, Y, A	110 × 110 × 300	1.8 kg	
WMF 025	140 CFM	238	1" BSP	P, X, Y, A	110 × 110 × 380	2.2 kg	
WMF 035	200 CFM	340	1.25" BSP	P, X, Y, A	140 × 140 × 440	3.6 kg	
WMF 060	350 CFM	595	1.5" BSP	P, X, Y, A	140 × 140 × 520	4.2 kg	

Key Engineering Advantages

- **Four-Stage Progressive Filtration Architecture:** Matches filtration precision exactly to process requirements: Grade P (3µm / 5 ppm oil), Grade X (1µm...
- **Alocrom Passivated Die-Cast Aluminum Housings:** Housings are pressure die-cast from high-grade aluminum, treated with Alocrom anti-corrosion chemical...

Primary Applications

- **Pharmaceutical Cleanrooms & Pill Packaging:** Guarantees zero oil contamination in sterile air lines compliant with FDA Class 1 com...
- **Food, Beverage & Dairy Processing:** Eliminates foul hydrocarbon odors and oil taste from compressed air contacting food c...
- **Automated Spray Painting & Powder Coating:** Stops micro-droplets of oil from creating craters, blisters, and surface blemishes on...

CONDENSATE MANAGEMENT & ENERGY CONSERVATION

Industrial Automatic Condensate Drain Valves

Electronic Timer Drains, Capacitive Zero-Air-Loss Drains & 40-Bar High Pressure Drain Systems for Guaranteed Condensate Evacuation Without Energy Waste



Win Equipments WADV Series automatic condensate drain valves eliminate moisture and oil slurry accumulated in compressor aftercoolers, air receivers, refrigerated dryers, and drop legs. Condensate buildup damages pneumatic cylinders and contaminates finished goods, while manual draining is frequently neglected. WADV valves provide fully automated, trouble-free drainage across three core technologies: electronic analog timer valves, intelligent capacitive zero-air-loss drains, and 40-bar high-pressure drain systems.

Working Principle: In WADV-T16 timer valves, a solid-state microcontroller energizes a direct-acting brass solenoid valve at preset intervals (0.5 to 10 seconds ON time; 0.5 to 45 minutes OFF time).

Design Ratings: Supply Voltage: 220V AC / 50Hz (Optional 24V DC / 110V AC) | Working Pressure: 0 to 16 bar g (HP model up to 40 bar g) | Ambient Temp: 1°C to 60°C | Enclosure: IP65 Heavy Duty | Body: Forged Brass / Stainless Steel

MODEL CODE	DRAIN TECHNOLOGY	OPERATING PRESSURE	IN/OUT CONN	DISCHARGE CYCLE	BODY METALLURGY	POWER SUPPLY
WADV-T16	Electronic Timer Drain	0–16 bar g	1/2" BSP In / Out	0.5–10s ON / 0.5–45m OFF	Forged Brass + SS Mesh	220V / 1Ph / 50Hz
WADV-Z16	Capacitive Zero-Air-Loss	0.8–16 bar g	1/2" BSP In / 1/4" Out	Demand-based (0 CFM Loss)	Die-Cast Aluminum	220V / 1Ph / 50Hz
WADV-HP40	High Pressure Timer	0–40 bar g	1/2" BSP High-Press	0.5–10s ON / 0.5–45m OFF	Heavy Stainless Steel	220V / 1Ph / 50Hz
WADV-Z40	Zero-Loss High-Pressure	1.2–40 bar g	1/2" BSP High-Press	Demand-based (0 CFM Loss)	Stainless Steel Body	220V / 1Ph / 50Hz

Key Engineering Advantages

- **Capacitive Zero Air Loss Technology (WADV-Z16):** Non-contact electronic capacitive sensor senses liquid level through the chamber wall with zero movi...
- **Built-In Stainless Steel Mesh Strainer:** WADV-T16 features a forged brass ball-valve strainer integrated directly into the valve body. Captur...

Primary Applications

- **Air Compressor Aftercooler Moisture Draining:** Instantly purges heavy condensate slugs discharged from compressor aftercoolers befor...
- **Air Receiver Bottom Sump Drainage:** Eliminates corrosive water pooling in receiver vessel dished bottoms, preventing inte...
- **Refrigerated & Desiccant Dryer Separators:** Discharges condensed water from 3-in-1 heat exchangers without wasting precious compr...

CROSS-INDUSTRY ENGINEERING SOLUTIONS

Industry Application Matrix & System Integration

Recommended equipment combinations across key industrial manufacturing sectors

INDUSTRY SECTOR	RECOMMENDED AIR DRYERS	PROCESS CHILLERS	COOLING TOWERS	PNEUMATIC AUXILIARIES
Textile Spinning & Weaving	WRD Series (200–2,000 CFM)	WCP Series (Central Chilled)	WCT-RL (100–500 TR)	WRV 1,000–5,000L + WADV-Z16
Plastic Injection & Extrusion	WRD Series (60–500 CFM)	WCP Series (5–50 TR, ±0.5°C)	WCT-RL (20–200 TR)	WADV-Z16 Capacitive Drains
Automotive Clear-Coat Paint	WHD Desiccant (-40°C PDP)	WCP / WCC Series	WCC Closed-Loop Towers	WMF Grade Y+A Microfilters
Pharmaceutical Formulation	WHD Series (-40°C / -70°C)	WCP / WAC Acid Units	WCT-SL Modular Towers	SUS304 Vessels + Grade A
Steel & Induction Furnaces	WRD Series	WCP Heavy Process	WCC Closed Serpentine	WRV Heavy Storage Vessels
Commercial Fisheries & Marine	WRD Series	WCP Chilled Brine Units	WCT Condenser Towers	WFI Flake Ice (1–30 TPD)
Hospital Diagnostic Imaging	—	WMS Series (N+1 Dual)	WCT Closed Circuit	City Water Auto-Bypass Skid
Fiber Laser Cutting & Optics	WRD / WHD Assist Gas	WCP Dual Circuit (+22°C/28°C)	—	WMF 0.01μ Filters + 40-Bar WRV

TURNKEY COMPRESSOR ROOM & UTILITY PIPING ENGINEERING

Win Equipments provides turnkey design, supply, and erection of compressed air ring-mains, stainless process water loops, and cooling tower circulating circuits. Sized to maintain flow velocities <6 m/s, eliminating line friction and pressure drops across extensive manufacturing halls.

DIRECT FACTORY FIELD ENGINEERING

Pan-India Service Network & Spares Depot

24/7 technical assistance, scheduled lifecycle AMC's, and immediate OEM spare parts availability



Coimbatore Works HQ & Central Depot

Central spare parts hub at Arasur with over ₹50 Lakhs in ready stock: Danfoss expansion valves, Copeland scroll compressors, activated alumina, and filter cartridges for immediate 24h dispatch.



Chennai & Sriperumbudur Automotive Hub

Dedicated technical service engineers stationed in Chennai supporting tier-1 automotive suppliers, robotic assembly lines, electronics factories, and seaport logistics complexes.



Bangalore & Hosur Precision Tooling Corridor

Rapid-response teams supporting precision CNC aerospace machine shops, medical device manufacturers, and high-power fiber laser fabrication facilities across Karnataka.



Tirupur & Erode Textile Industrial Belt

Same-day breakdown attendance and scheduled preventive maintenance for spinning mills, garment dyeing plants, and knitwear processing houses throughout Western Tamil Nadu.



Comprehensive AMC Contracts

- Quarterly thermodynamic health and refrigerant charge audits
- Ultrasonic compressed air leak surveys to identify energy loss
- Desiccant bed moisture testing and dew point logging
- Priority 4-hour breakdown response guarantee

⚙️ Genuine OEM Consumables Hub

- Spherical Activated Alumina (3–5mm & 4–8mm) in 25/50 kg bags
- Molecular Sieve 13X / 4A for deep -70°C dew point drying
- Replacement filter cartridges compatible with all major housings
- Synthetic compressor and chiller refrigeration oils

GLOBAL ENGINEERING INQUIRIES • DIRECT FACTORY CONNECT

Request an Engineering Consultation & Custom Sizing

Our thermodynamic application specialists analyze your flow rates, ambient conditions, and temperature requirements to engineer the most cost-effective and energy-efficient solution.

WIN EQUIPMENTS

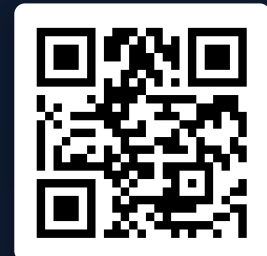
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📍 SF No: 4, 195 B, Kallangadu, Arasur Post, Coimbatore – 641407, Tamil Nadu, India

☎ **+91 95972 28969 • +91 95972 28975**

✉ info@winequipments.com • winequipments.com

GPS Plant Coordinates: Latitude 11.0507° N, Longitude 77.1084° E (NH544 Corridor)



SCAN FOR WEB PORTAL